

Workflow

Hospital Emergency Room Dashboard Project

1. Problem Understanding

The project began by identifying the business problem and defining the objectives of the analysis. The objective was to develop an interactive Power BI dashboard that transforms historical Emergency Room (ER) data into meaningful insights for operational monitoring and data-driven decision-making.

2. Data Understanding

The dataset was explored to understand its structure, variables, and quality. Key fields included Patient ID, Admission Date, Patient Age, Gender, Race, Department Referral, Patient Wait Time, Patient Satisfaction Score, and Admission Status. Data types, missing values, duplicate records, and variable distributions were examined.

3. Data Cleaning and Preprocessing

The dataset was prepared by verifying data types, retaining Patient ID as text because it contains hyphens, converting Admission Date into Date/Time format, checking for duplicates and missing values, cleaning text fields, and creating Admission Year, Admission Month, Admission Month Number, Admission Day, Admission Hour, Age Group, and Age Group Sort.

4. Data Modeling

A flat data model was adopted because the dataset consisted of a single structured table containing all variables required for analysis. No relationships between tables were necessary, making the model suitable for descriptive analytics and dashboard reporting.

5. Exploratory Data Analysis (EDA)

Exploratory analysis identified trends and patterns including total patients, total admissions, average waiting time, average satisfaction score, monthly ER visits, hourly arrivals, patient demographics, and department referrals.

6. Dashboard Development

An interactive Power BI dashboard was developed with KPI cards, trend charts, demographic visualizations, department referral analysis, interactive slicers, and a Key Insights panel.

7. Key Insights and Findings

The dashboard summarized patient volume, admissions, waiting time, satisfaction, monthly and hourly trends, referral patterns, and demographic distributions to support operational understanding.

8. Recommendations

Recommendations included increasing staffing during peak hours, improving patient flow, monitoring patient satisfaction, using historical trends for workforce planning, strengthening referral coordination, and continuing the use of business intelligence dashboards.

9. Conclusion

The project demonstrated how descriptive analytics and Power BI can transform historical healthcare data into actionable insights that support evidence-based decision-making and operational improvement.